

Niayesh Afshordi

Astrophysics, cosmology, and gravity

Professor, University of Waterloo; Research Associate Faculty,
Perimeter Institute
nafshordi@pitp.ca
nafshordi.com

Short public CV — updated 11 August 2026

Profile

Niayesh Afshordi is an astrophysicist whose research connects fundamental questions in gravity and quantum physics to observations of the universe. His work spans black holes, gravitational waves, the early universe, the cosmic microwave background, dark matter and dark energy, cosmic structure, and observational tests of gravity, with interdisciplinary applications in medical physics and epidemiological modelling.

Appointments

Full Professor, Department of Physics and Astronomy, University of Waterloo	2022–present
Research Associate Faculty, Perimeter Institute for Theoretical Physics	2009–present
Founding Member, Waterloo Centre for Astrophysics	2018–present

Education and previous appointments

PhD, Astrophysics, Princeton University	2000–2004
BSc, Physics, Sharif University of Technology	1996–1999
Distinguished Research Fellow, Perimeter Institute	2007–2009
Postdoctoral Fellow, Institute for Theory and Computation, Harvard–Smithsonian CfA	2004–2007

Selected research support

NSERC Discovery Grant: “Quantum Black Holes, Vacuum, and the Big Bang”	C\$250,000 2025–2030
University of Waterloo Interdisciplinary Trailblazer Grant: “Quantum Spacetime in the Cosmos”	C\$80,000 2022–2025
NSERC Discovery Grant: “Novel Astrophysical Windows into Fundamental Physics”	C\$170,000 2020–2025

Consulting

Aligned — consulting on testing frontier AI models in scientific research	March 2026–present
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Selected recognitions

Buchalter Cosmology Prize	2019; 2015
Professor M. K. Vainu Bappu Gold Medal	2011
Ontario Early Researcher Award	2011
NSERC Discovery Accelerator Supplement	2010–2013

Research group and scholarly record

Afshordi supervises students and postdoctoral researchers across cosmology, gravity, quantum field theory in curved spacetime, and data-informed astrophysics. A current, public record of papers, talks, group members, and writing is maintained at nafshordi.com. Complete publication listings are available through [arXiv](https://arxiv.org), [INSPIRE](https://inspirehep.net), [ORCID](https://orcid.org), and [Google Scholar](https://scholar.google.com).